

Anas TBER

Data Science & AI Engineering Student — Centrale Lyon

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Engineering student at Centrale Lyon with hands-on experience in ML, NLP, LLM applications and large-scale data engineering. Seeking an **end-of-studies internship in Data Science or AI** starting **March 2027**.

Education

- École Centrale de Lyon** — *General Engineering* *Lyon · Sep. 2023 – Mar. 2027*
- **Courses:** Machine Learning & Applications, Collaborative Algorithms, Probability & Stochastic Processes, Algorithms for Reasoning, OOP, Enterprise Management
- Lycée Henri Loritz** — *CPGE, PCSI / PSI** *Nancy · Sep. 2020 – Jun. 2023*
- Intensive preparatory classes in Physics, Mathematics & Engineering Sciences for France's top engineering school entrance exams (Grandes Écoles).

Experience

- Data Scientist Intern — BNP Paribas CIB** — *Compliance, Anti-Money Laundering* *Paris · Jan. 2026 – Jul. 2026*
(AML)
- Delivered a **regulator-approved** network-based AML detection & investigation platform on a **10M+ entity / 40M+ transaction** dataset, from raw ingestion to compliance dashboards; built the pipeline in **Polars / PyArrow** lazy mode with watchlist enrichment
 - Designed the detection layer on engineered graph features (centrality, pass-through ratio, flagged-neighbour ratio): **cycle detection** (SCC + DFS) to surface money-mule rings, and an **Isolation Forest** model with **SHAP** explainability ranking suspicious communities with regulator-facing justifications
 - Shipped **LLM-powered** features into the **Streamlit** app: one-click AI risk summaries of a community graph and a multi-turn **investigation chatbot** answering free-form questions on an account and its 3-hop neighbourhood (prompt engineering, context-window management)
 - **Sole owner and maintainer of the repository** — led the refactor (modular architecture, centralized configuration) and wrote the **full technical documentation** enabling handover to the IT team
- Tech & Operations Intern — Gentle Mates** — *Professional Esports* *Paris · Apr. 2025 – Nov. 2025*
- Built an automated social media analytics dashboard (Streamlit/Python) integrating computer vision (**YOLO**) for sponsor logo detection and NLP (**Regex**) for brand mention tracking, improving partnership ROI visibility
 - Contributed to the development of impactful software and AI internal tools for analytics, monitoring, and operations

Academic & Personal Projects

- YouTube Sentiment Analyzer** — *Personal Project* *2026*
- **Streamlit** web app classifying YouTube comment sentiment with **Hugging Face Transformers (RoBERTa, multi-lingual BERT)**; ingests up to 1,000 comments per video via the **YouTube Data API v3** and renders the breakdown through interactive **Plotly** dashboards
- Transformer Architectures & NLP** — *Research Project* *2024*
- Studied attention mechanisms, Word2Vec/GloVe/BERT embeddings, and Transformer vs. **Mamba** (SSM) complexity trade-offs
- Ant Colony Optimization — Shortest Path Solver** *2024*
- Implemented **ACO** metaheuristic in Python (OOP) with pheromone trails and probabilistic decision-making; built a real-time **Tkinter** visualization of pathfinding convergence

Skills

Programming	Python, JavaScript
AI & Data Science	Pandas, NumPy, Scikit-Learn, PyTorch, LightGBM, SHAP, LangChain, Jupyter
LLM & NLP	Prompt engineering, RAG-style context injection, Anthropic/OpenAI APIs, Transformers
Data Engineering	Polars, PyArrow, NetworkX, igraph, SQL, PostgreSQL
Web & Viz	Streamlit, Pyvis, Plotly, React, HTML/CSS
DevOps	Git, GitHub, GitLab
Languages	French (Native), Arabic (Native), English (Fluent), Spanish (Beginner)